

CUSUM Change-Point Detection Analysis

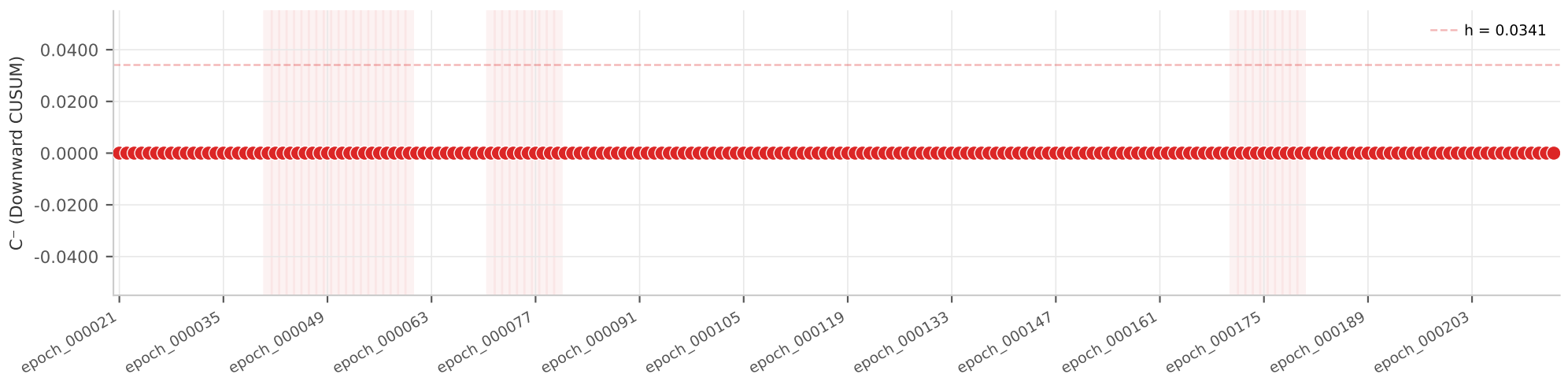
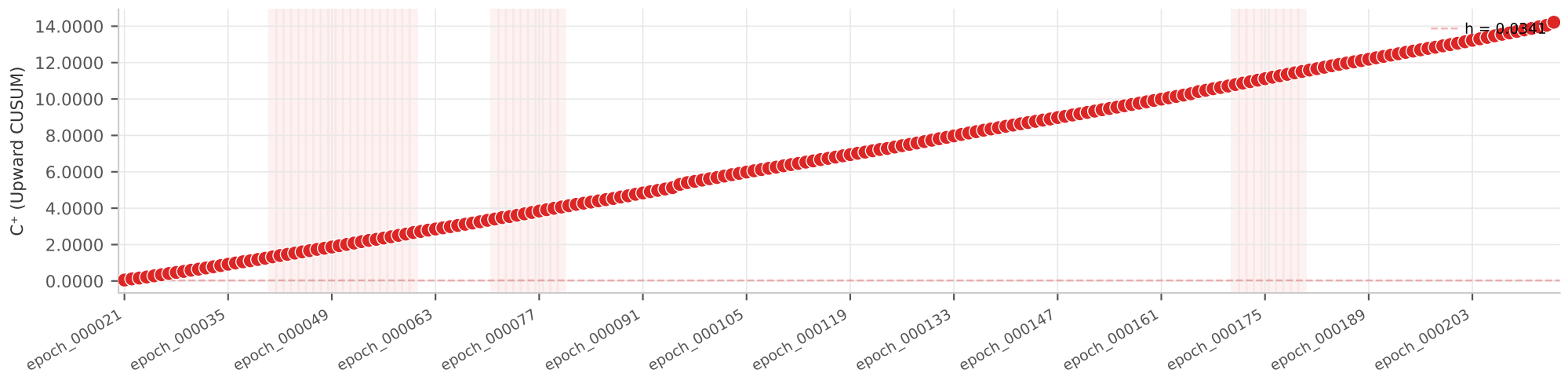
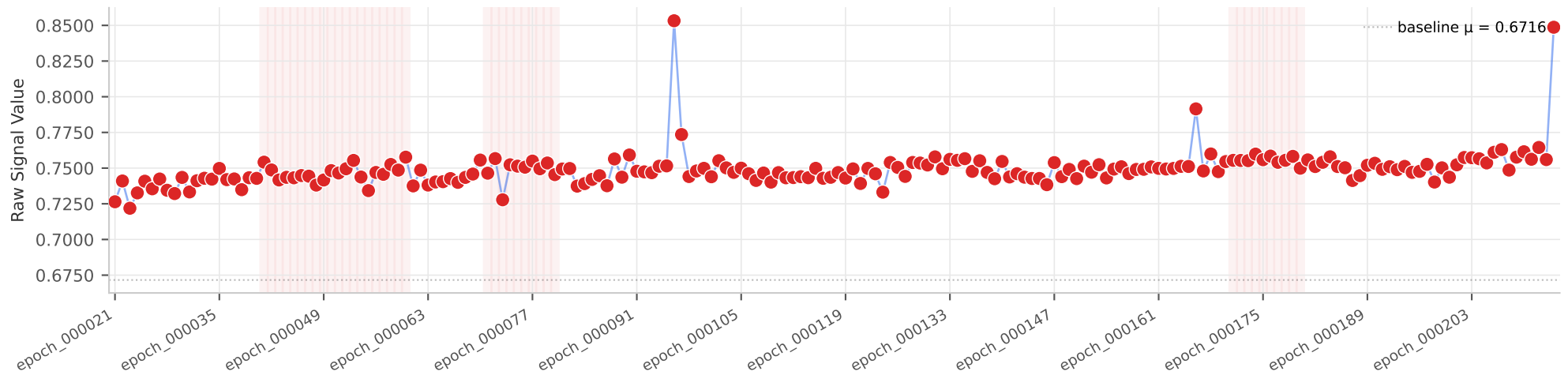
Cumulative Sum Control Chart | JS Divergence + New IP Ratio + Top-K Source

Run Parameters		Baseline CUSUM Parameters		Detection Summary (Any Signal)	
Dataset	txtD280_10.0_5.0_cusum_sparse.json	Baseline JS	$\mu=0.6716$ $\sigma=0.0085$ $k=0.0043$ $h=0.0341$	True Positives (TP)	40
Timestamp	2026-02-23T14:52:08.790605	New IP Ratio	$\mu=0.0000$ $\sigma=0.0000$ $k=0.0000$ $h=0.0000$	True Negatives (TN)	154
Method	CUSUM Change-Point (JS + New IP + TopK)	Top-K	$\mu=0.0623$ $\sigma=0.0024$ $k=0.0012$ $h=0.0096$	False Positives (FP)	0
Total epochs	—	False Negatives (FN)			0
Baseline epochs	— (?%)	Precision			0.206
Attack epochs	194	Recall			1.000
Malicious traffic	?%	F1 Score			0.342
CUSUM k-factor	0.5	Detection Rate			100.0%
CUSUM h-factor	4.0				
Mode	SPARSE				
Attack probability	0.3				
Window size	10				
Windows attacked	4				

Combined JS Divergence

CUSUM Signal Analysis | malicious = 0% | baseline = 5%

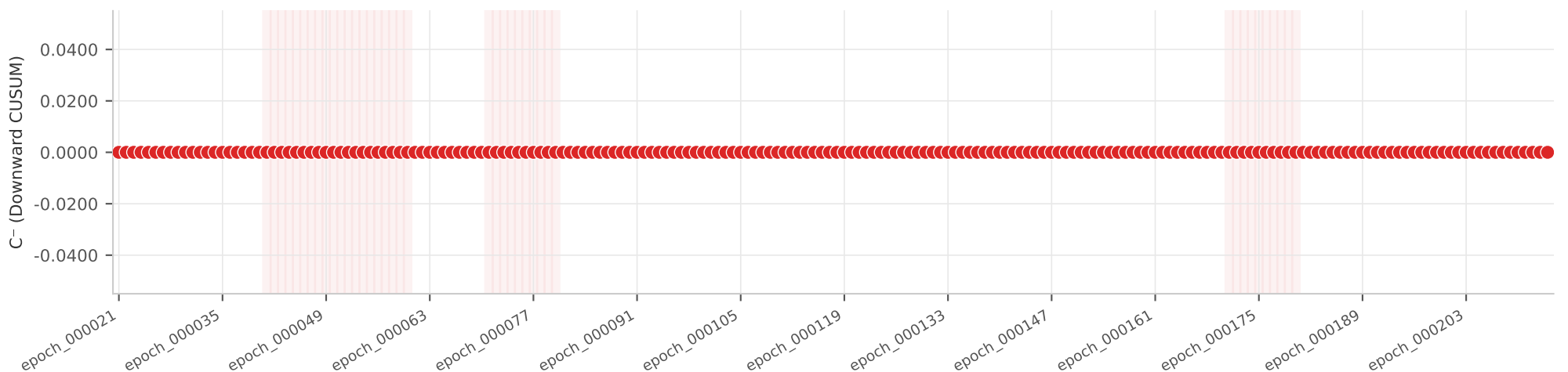
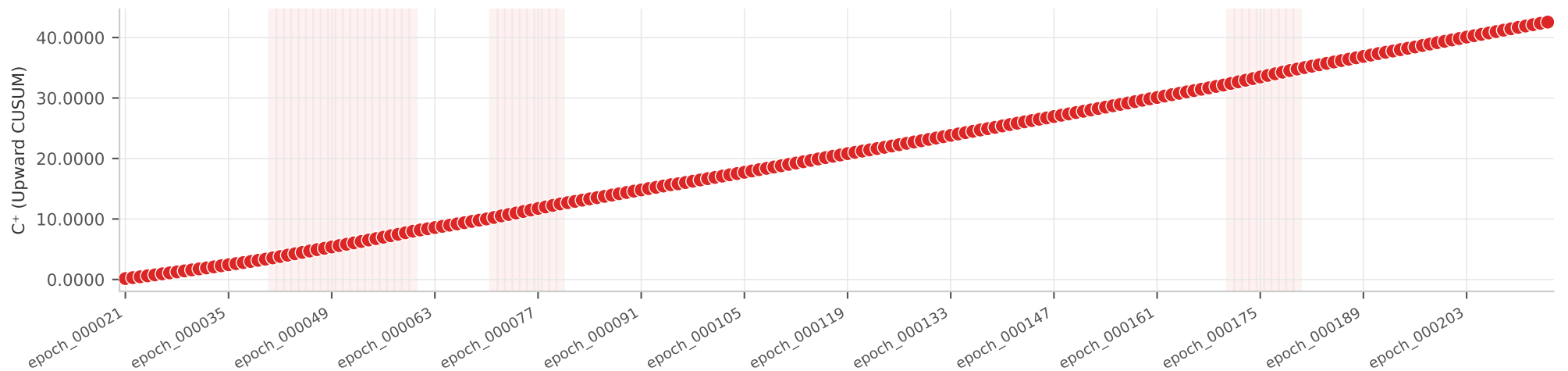
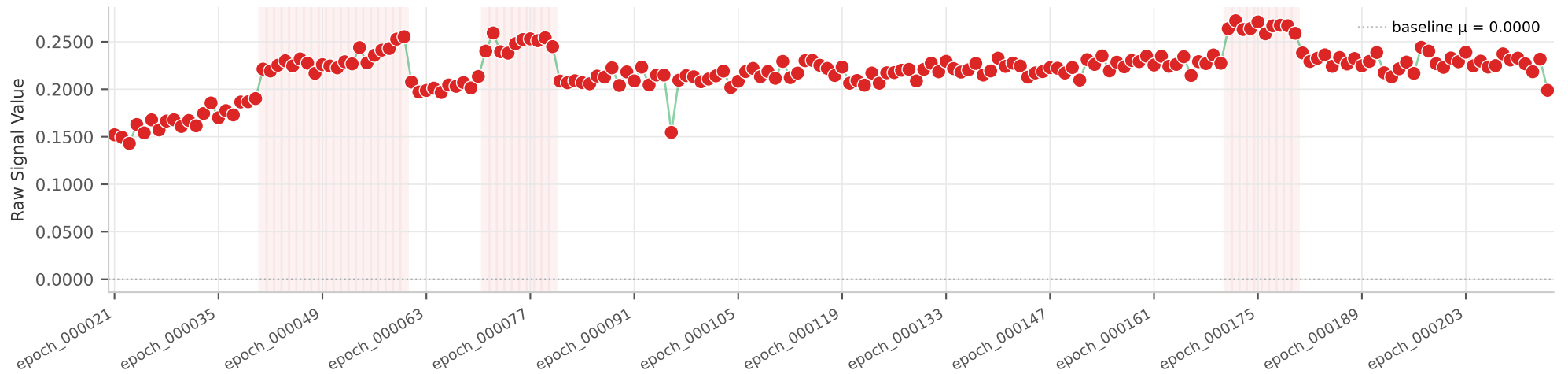
- CUSUM alarm fired
- Attacked epoch



New Source IP Ratio

CUSUM Signal Analysis | malicious = 0% | baseline = 5%

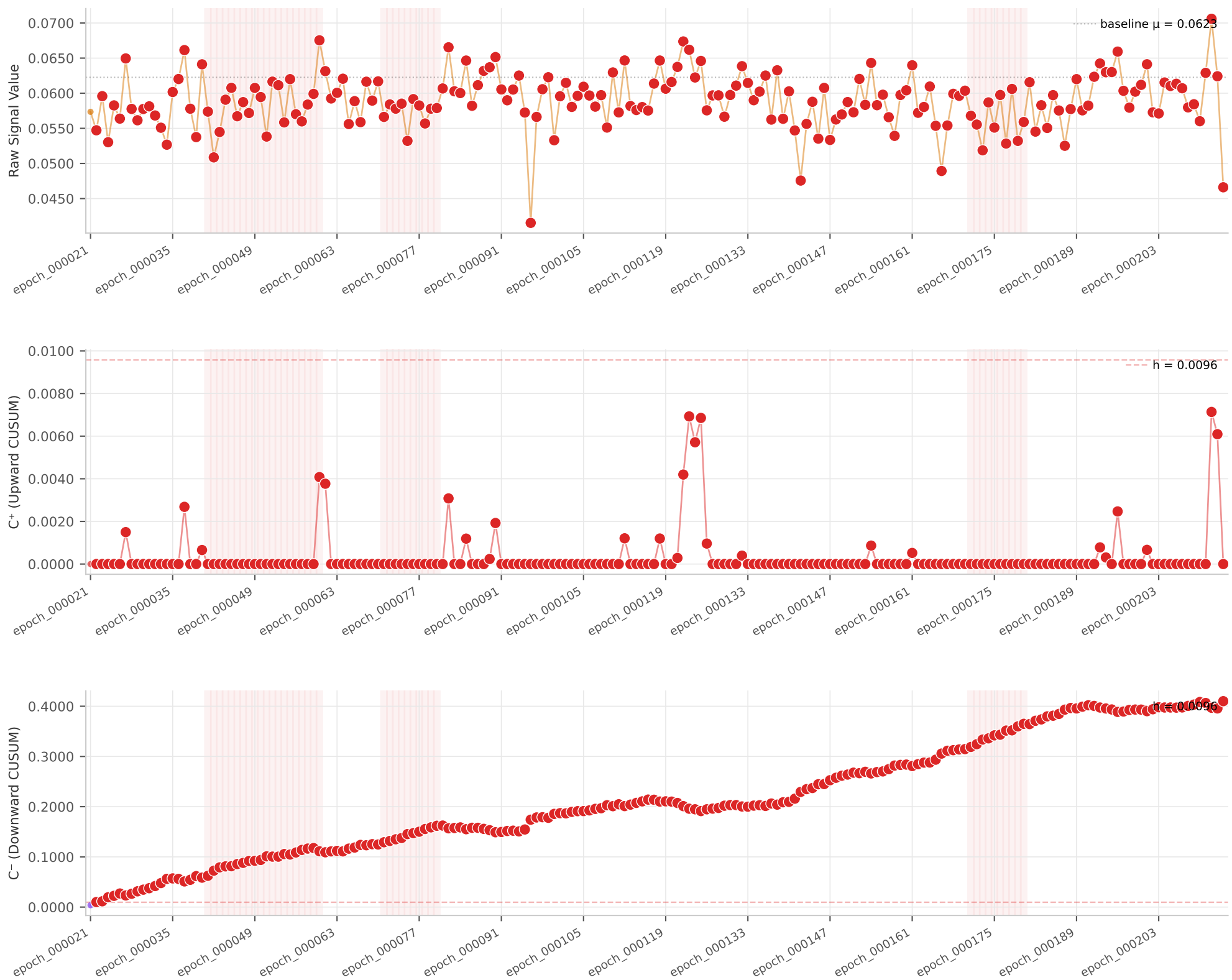
- CUSUM alarm fired
- Attacked epoch



Top-5 Source Concentration

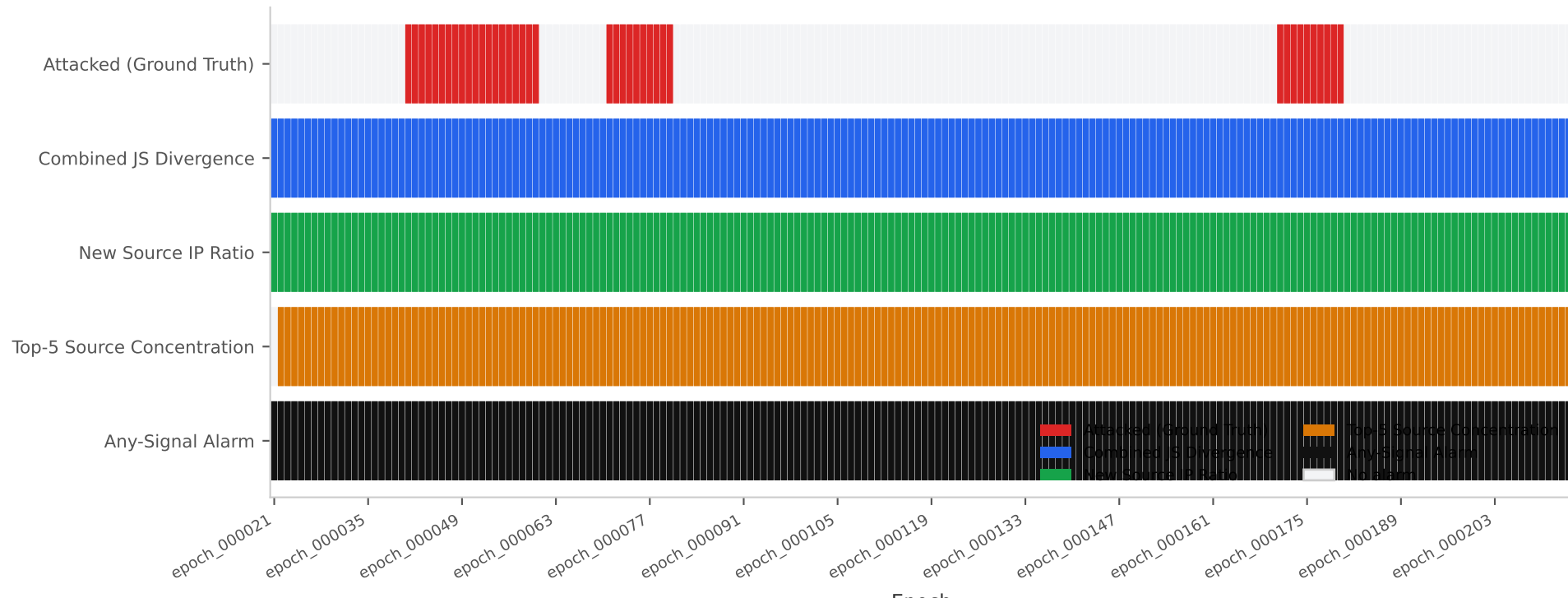
CUSUM Signal Analysis | malicious = 0% | baseline = 5%

- CUSUM alarm fired
- Attacked epoch



CUSUM Alarm Timeline

Filled cell = alarm fired for that epoch × signal



Detection Performance Metrics

Dataset: txtD280 | Malicious: ?% | Baseline: ?%

Metric	Value	Description
True Positives	40	Attacked epochs correctly alarmed
False Positives	154	Clean epochs incorrectly alarmed
True Negatives	0	Clean epochs correctly not alarmed
False Negatives	0	Attacked epochs missed
Precision	0.2062	$TP / (TP + FP)$
Recall	1.0000	$TP / (TP + FN)$
F1 Score	0.3419	Harmonic mean of Precision & Recall
Detection Rate	100.00%	Fraction of attacked epochs detected
False Alarm Rate	100.00%	Fraction of clean epochs with alarm